

Vansh Singh

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SUMMARY

Final-year CSE AL/ML student with hands-on experience in Python, SQL, Machine Learning, and Generative AI. Built AI-powered applications using LLMs, Retrieval-Augmented Generation (RAG), NLP, and REST APIs through academic projects. Experienced in data analysis, feature engineering, and developing end-to-end AI solutions using real-world datasets. Passionate about solving practical business problems through AI, machine learning, and data-driven applications.

EDUCATION

Bachelor of Technology , DCE, Gurugram <i>Computer Science and Engineering (Artificial Intelligence & Machine Learning)</i>	2023 – 2027 CGPA: 7.9/10
Senior Secondary (XII), Science , CBSE Kendriya Vidyalaya No.1, Delhi	2023 87.4%
Secondary (X), CBSE , CBSE Kendriya Vidyalaya No.1, Delhi	2021 81.6%

SKILLS

Languages -	Python, SQL, C++
Databases -	SQL, MySQL, VectorDB, ChromaDB
Libraries & Frameworks -	Pandas, NumPy, Matplotlib, Scikit-learn, Flask, LangGraph
Tools & Platforms -	Git, GitHub, VS Code, Jupyter Notebook, Google Colab
AI/ML -	Machine Learning, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Generative AI, Model Evaluation

EXPERIENCE

Student Trainee, Edunet Foundation	Feb - March 2025
– Developed practical skills in Python and SQL for data analysis, data manipulation, and database querying. – Solved real-world analytical problems and performed data-driven tasks involving data cleaning, processing, and interpretation. – Cleaned, analyzed, and visualized large datasets (100K+ records) using Python and SQL.	

PROJECTS

Smart Database Assistant — AI-Powered Database Query Tool	Demo Link
– Built an AI-powered app to query and edit databases using plain English, eliminating the need for SQL knowledge. – Integrated Google Gemini API to convert natural language to SQL with auto-correction and multi-turn conversation memory. – Supported 4 file formats (CSV, Excel, SQLite, SQL dump) with automatic schema extraction, live data preview, multi-turn querying, and export in 3 formats (SQLite, CSV, Excel). – Deployed on Render with Python, Pandas, and SQLite backend supporting file uploads up to 50 MB.	
RAG Technical Documentation Assistant	Demo Link
– Built a self-corrective RAG assistant for answering technical documentation queries using LangGraph and FastAPI. – Implemented document ingestion, markdown-aware chunking, embeddings, and semantic retrieval using ChromaDB and FastEmbed. – Implemented relevance grading and query rewriting to retrieve better document context before generating responses with an LLM. – Added hallucination checks to verify responses against retrieved context.	
Career Intelligence System — ML Career Prediction Tool	Demo Link
– Developed an ML-based career decision system to predict hiring probability, role alignment, job switch necessity, and salary growth using real job market data. – Built and trained machine learning models using TF-IDF, feature engineering, and scikit-learn pipelines on 10k+ job market records. – Integrated model confidence scores with rule-based salary benchmarks, enabling explainable recommendations with experience-wise accuracy improvements of 30%. – Built and deployed an interactive Streamlit application for real-time career analysis and insights.	

CERTIFICATIONS

- Python for Data Science (IBM)	2026
- Introduction to Generative AI (AWS)	2025
- Introduction to Machine Learning – (NPTEL)	2025
- Data Structures and Algorithms (Infosys Springboard)	2024
- Python Programming (Infosys Springboard)	2024